

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

L23

Sep 18, 2026

Ang and Tang

1. CLT Section 4.2.2
2. Extreme Value Distributions 4.2.3

Messages from the Last Class

• Message 1: Central Limit Theorem (CLT)

→ Applies to means

→ Always follows "Normal"

→ with mean same as population (μ)

→ with stdev. less than that of population ($\frac{\sigma}{\sqrt{n}}$)

n : number of Random Variables being Averaged.

Two mechanisms

: Averaging: Depends of # of RVs being averaged

: Repetitions: Performed to obtain smooth PDF/CDF

→ Averaging is the ONLY mechanism that reduces dispersion

Announcement

- On GitHub repo, added a new directory
 - `class-demos\M1C4-clt-demo.xlsx`
 - `class-demos\M1C4_gev.py`
- Check Moodle regularly (courses.iitm.ac.in)
 - *Yet to upload*
 - *the course outline*
 - *notes from the first three lectures*

Gumbel distributions

- When we are interested in

smallest

or

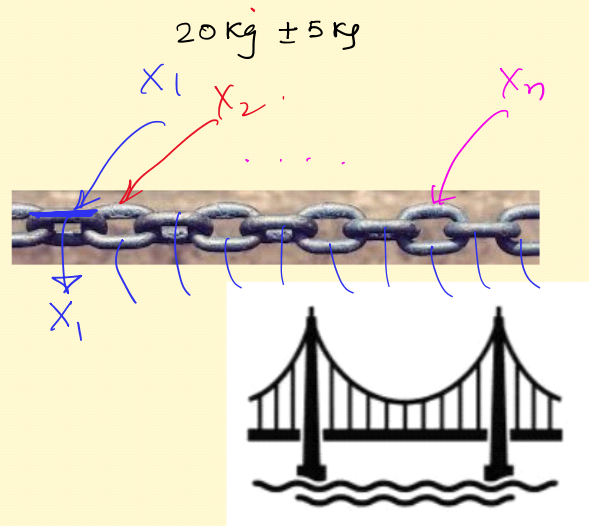
Largest of several RVs.

minimum expense

$$Y_1 = \min(X_1, X_2, \dots, X_n)$$

maximum expense

$$Y_n = \max(X_1, X_2, \dots, X_n)$$



5

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$

$$Y_n = \max(X_1, X_2, \dots, X_n).$$

$$F_X(x)$$

X_i 's $\rightarrow$ i.i.d.
independent
identical

The CDF of Y_1 is (smallest extreme):

$$F_{Y_1}(y) = \Pr(Y_1 \leq y) = \Pr[\min(X_1, X_2, \dots, X_n) \leq y]$$

$$= 1 - \Pr[\min(X_1, X_2, \dots, X_n) > y]$$

$$= 1 - \Pr[(X_1 > y) \cap (X_2 > y) \cap \dots \cap (X_n > y)]$$

$$= 1 - \prod_{i=1}^n \Pr(X_i > y)$$

CDF

PDF

$$\Pr(Y_1 \leq y) = 1 - [1 - \Pr(X \leq y)]^n$$

Gumbel distributions

$$Y_1 = \min(X_1, X_2, \dots, X_n),$$
$$Y_n = \max(X_1, X_2, \dots, X_n).$$

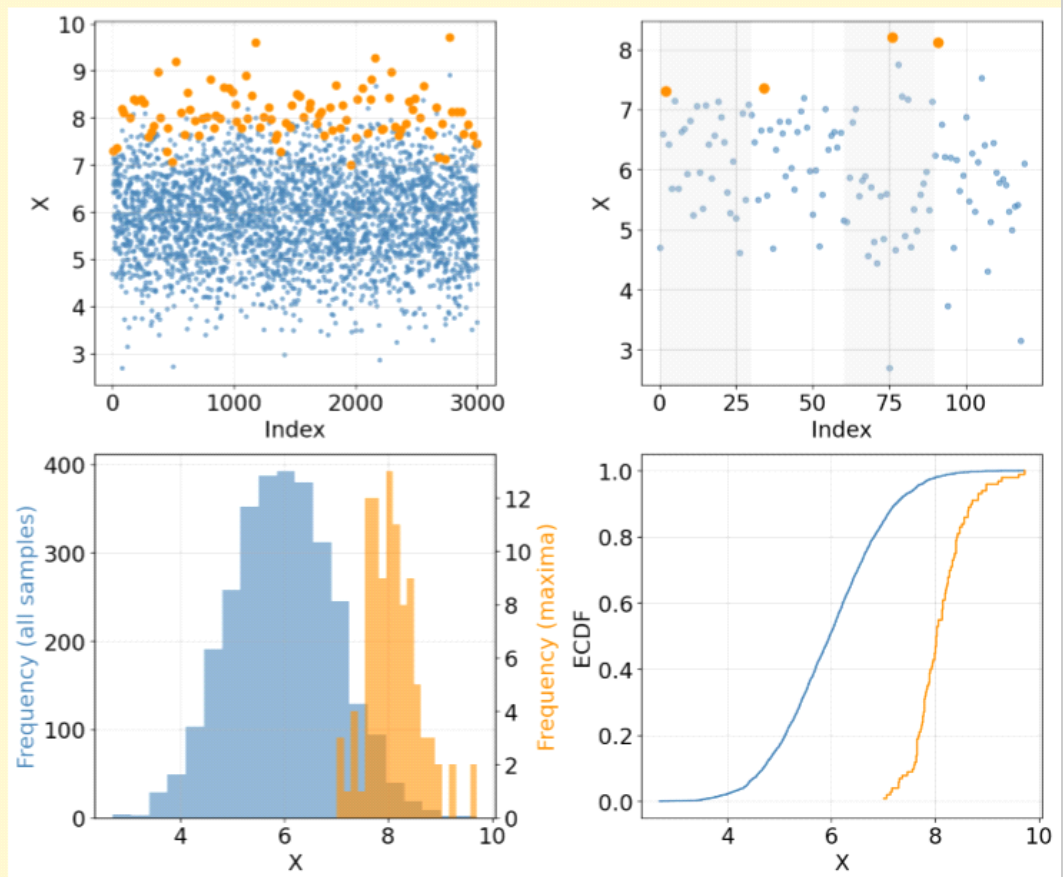
The CDF of Y_n is (largest extreme):

$$F_{Y_n}(y) =$$

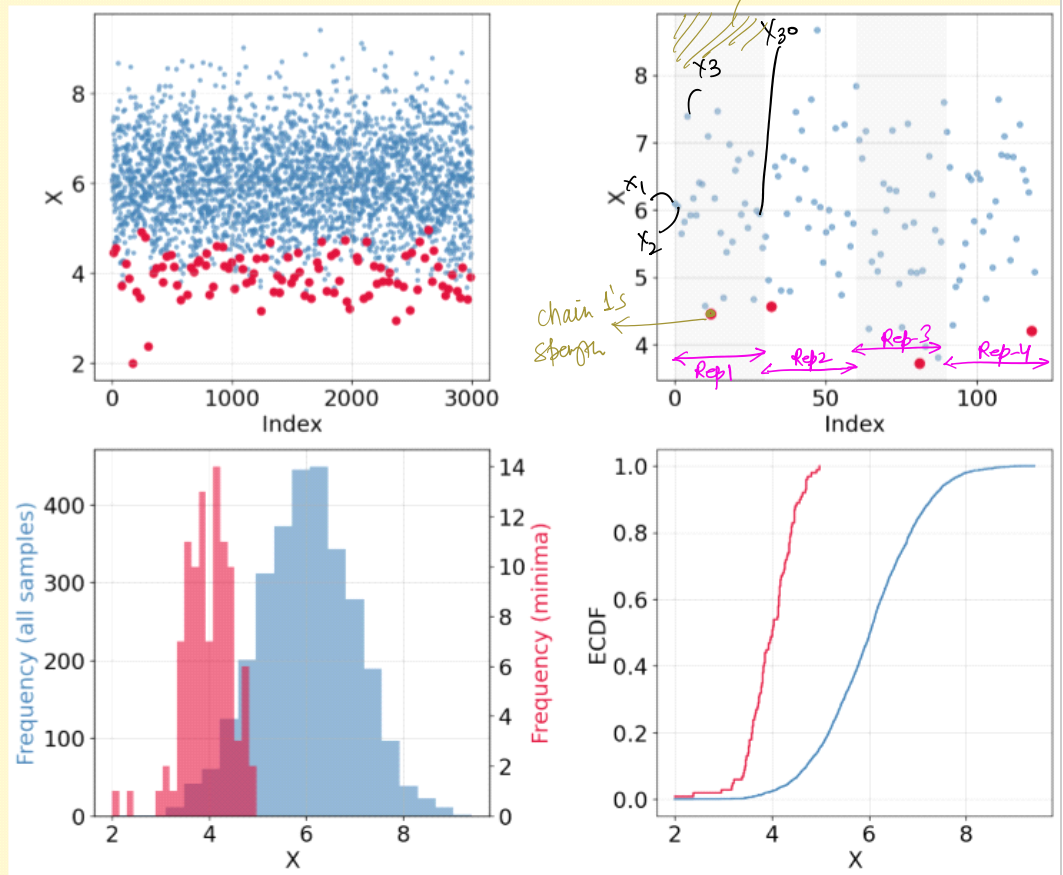
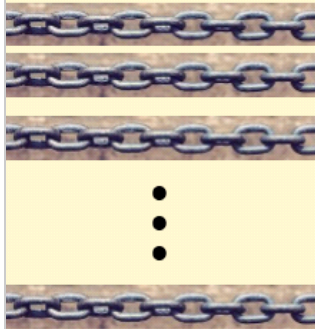
CDF

PDF

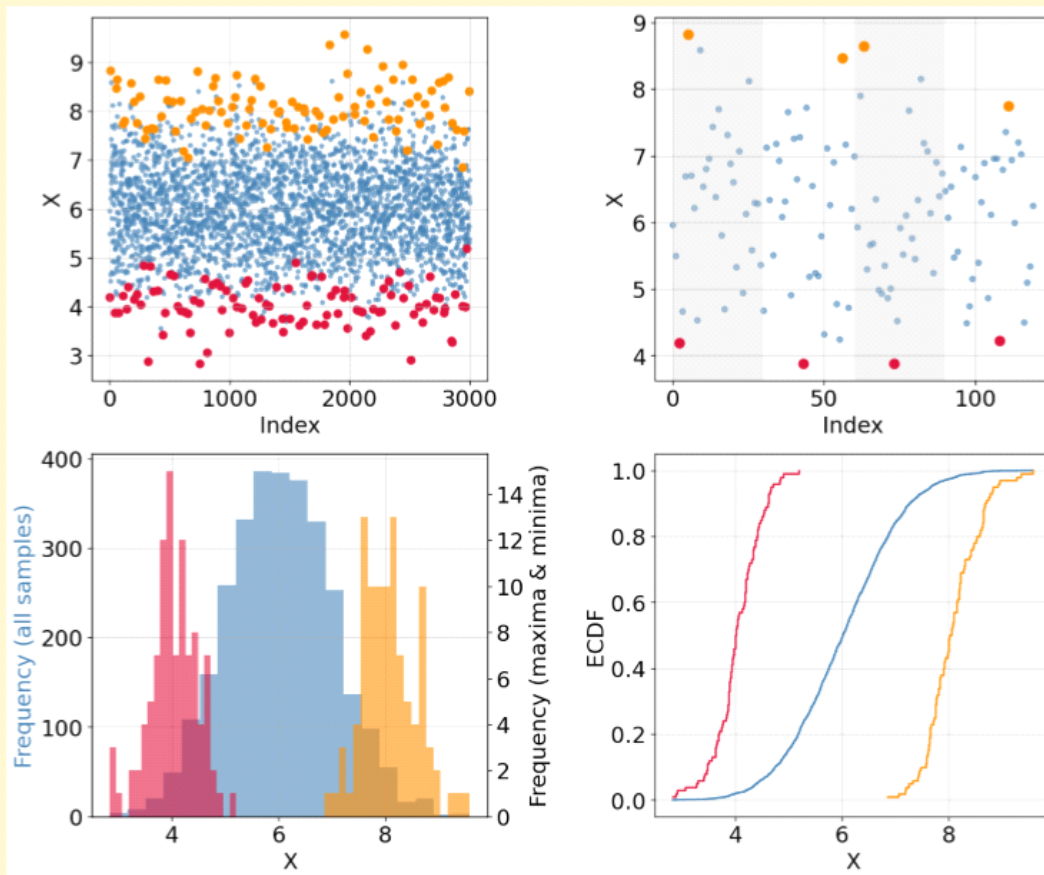
Extreme value analysis



Extreme value analysis

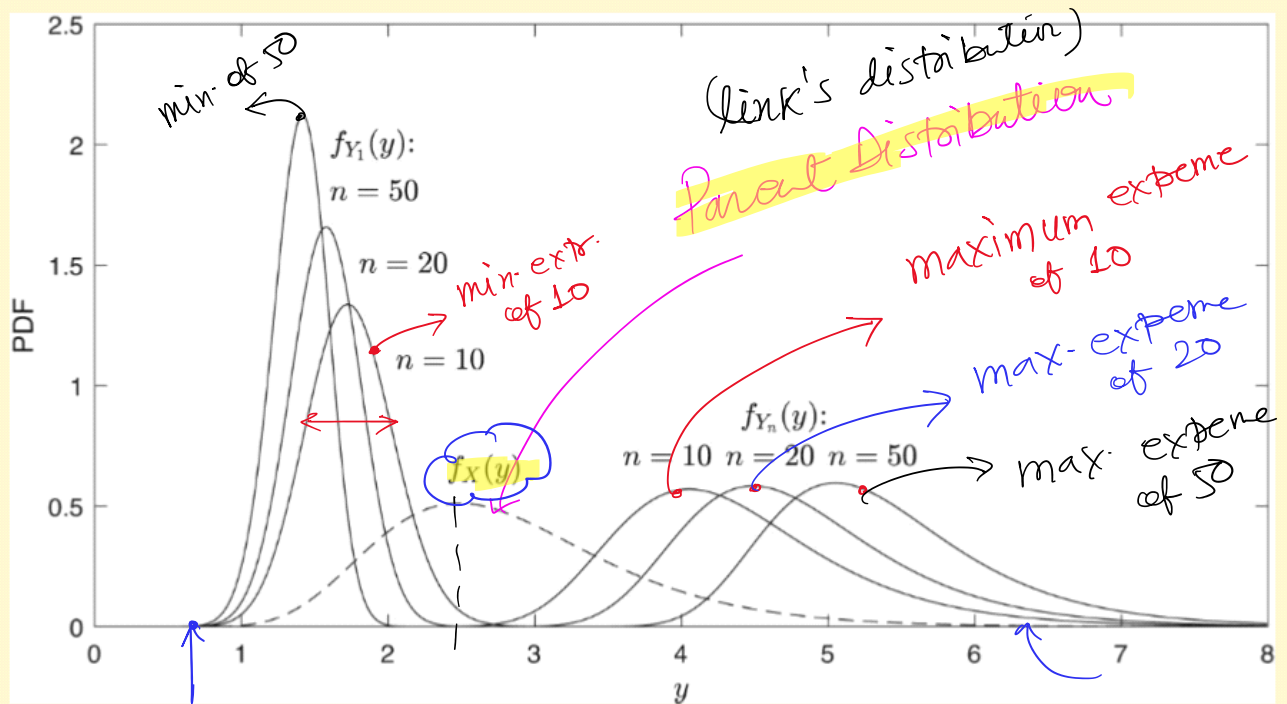


Extreme value analysis



10

Gumbel distributions



$$\sigma_{Y_1 \text{ for } n=10} > \sigma_{Y_1 \text{ for } n=50}$$